Resonance Folding: Lossless Octonion Weight Representation and Geodesic Model Merging for Native 8-Channel Convolutional Networks

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Abstract

We present **Resonance Folding** (RF), a weight representation technique that encodes convolutional filter weights as unit octonions on the unit 7-sphere S^7 using the Fano-plane algebra. For networks whose convolutional layers use channels divisible by 8—which we call *native 8-channel networks*—the RF encoding is provably lossless: reconstruction cosine similarity is 1.000000 and holographic coherence loss is 0.000000, regardless of training outcome. We introduce **OctConvNet**, a convolutional architecture family satisfying this property by design, and demonstrate that OctConvNet-M achieves 88.34% accuracy on CIFAR-10, exceeding a parameter-matched float32 baseline (88.15%) by $+0.19\%$. We show that geodesic interpolation via spherical linear interpolation (OCT-SLERP) on S^7 outperforms arithmetic weight averaging in all tested conditions: by $+41.98\%$ under complementary data-split merging, and by $+0.19\%$ over the best individual fine-tuned model on CIFAR-10. Across two seeds on CIFAR-10 and one run on CIFAR-100, OCT-SLERP outperforms the best individual model in every condition (mean $+0.23\% \pm 0.24\%$ on CIFAR-10; $+0.40\%$ on CIFAR-100). Holographic coherence is preserved exactly ($\mathcal{H} = 0.000000$) at every interpolation point across all experiments. Both properties are scale-invariant, holding identically at $4\times$ model scale (143,424 octonion kernel groups, ± 0.004 delta from OctConvNet-M). We further demonstrate lossless fold on OctResNet-18 (1.39M octonion groups, standard ResNet-18, no channel changes), and on SmolLM2-135M MLP blocks (9.95M groups, 90 layers, $\mathcal{H} = 7.96 \times 10^{-16}$). Partial-network OCT-SLERP on SmolLM2-135M outperforms float averaging by 0.40 perplexity points on WikiText-2. Sequential OCT-SLERP continual learning reduces catastrophic forgetting by 92% vs. naive fine-tuning and outperforms EWC [Kirkpatrick et al., 2017] without Fisher matrix computation. We release `resonance-folding`, a Python package implementing the full pipeline.

1 Introduction

Neural network model merging—combining the weights of independently trained models into a single model—has attracted significant interest as a means of aggregating complementary knowledge from models fine-tuned on different tasks or data distributions [Wortsman et al., 2022a, Ilharco et al., 2023, Yadav et al., 2023, Yu et al., 2023]. The dominant approach is arithmetic averaging in float space: the merged model's parameters are the element-wise mean of the source models' parameters.

We identify a fundamental geometric problem with this approach. When two models are trained on complementary data distributions, their weight vectors diverge in direction—in the extreme case occupying near-antipodal regions of weight space. Arithmetic averaging of such vectors produces a result near the origin: a model whose weights carry no coherent representation of either training distribution. In our experiments, float averaging under complementary-split conditions collapses to random-chance accuracy (10.0% on CIFAR-10, 0.99% on CIFAR-100).

The geometrically correct interpolation between two points on a sphere is *spherical linear interpolation* (SLERP), which follows the great-circle arc and produces intermediate points that remain on the sphere. For SLERP to be meaningful for neural network weights, those weights must live on a sphere.

We show this condition is satisfied naturally for native 8-channel convolutional networks. Each convolutional filter with 8 input channels is an 8-dimensional vector—a natural candidate for $S^7 \subset \mathbb{R}^8$. The Fano-plane octonion algebra defines multiplication between elements of $\mathbb{R}^8$ such that unit octonions satisfy $\mathbf{o} \cdot \mathbf{o}^\dagger = \mathbf{1}$ exactly. We call the projection of native 8-channel conv weights onto S^7 a *Resonance Fold*, and show this projection is lossless by construction.

Contributions.

1. RF: a lossless weight representation for convolutional networks with 8-aligned channel widths, projecting filter weights onto S^7. Standard ResNet-18 satisfies this property with no architecture modifications.
2. **OctConvNet**: a convolutional architecture family satisfying 8-channel alignment by design, with oct-aware weight initialization.
3. Empirical proof that RF fold is lossless across all tested scales and training runs ($\cos = 1.000000$, $\mathcal{H} = 0.000000$, $\Delta\text{acc} = 0.0000$).
4. OCT-SLERP merging outperforms float averaging under both complementary split ($+41.98\%$) and constructive divergence ($+0.19\%$ over best individual) conditions.
5. Scale invariance: both properties hold at $4\times$ model scale (± 0.004 delta).
6. Open-source Python package: `pip install resonance-folding`.
7. **SLERP continual learning**: sequential SLERP merging reduces catastrophic forgetting by 92% vs. naive fine-tuning and outperforms EWC [Kirkpatrick et al., 2017] while requiring no Fisher matrix computation.

2 Background

2.1 Octonion Algebra

Octonions are the largest normed division algebra over $\mathbb{R}$ (after reals, complex numbers, and quaternions). An octonion $\mathbf{o} \in \mathbb{R}^8$ has one real and seven imaginary components:

$$\mathbf{o} = e_0 + e_1 i + e_2 j + e_3 k + e_4 l + e_5 m + e_6 n + e_7 p. \tag{1}$$

Multiplication follows the Fano-plane convention (seven cyclic triples), producing a 64-term expansion. The algebra is non-commutative ($AB \neq BA$) and non-associative ($(AB)C \neq A(BC)$) but satisfies the alternative law.

The conjugate negates imaginary components: $\mathbf{o}^\dagger = e_0 - e_1 i - \cdots$. For a unit octonion ($\|\mathbf{o}\| = 1$):

$$\mathbf{o} \cdot \mathbf{o}^\dagger = \mathbf{1} \quad \text{(exactly)}. \tag{2}$$

Holographic coherence loss. We define a quality metric for a set of octonion shards $\mathbf{O} \in \mathbb{R}^{N \times 8}$:

$$\mathcal{H}(\mathbf{O}) = \frac{1}{8} \cdot \text{mean}\left(\left\| \mathbf{O} \cdot \mathbf{O}^\dagger - \mathbf{I} \right\|_F^2 \right). \tag{3}$$

For perfect unit octonions, $\mathcal{H} = 0$ exactly.

2.2 SLERP on S^7

For two unit octonions $\mathbf{A}, \mathbf{B} \in S^7$, the geodesic (great-circle arc) interpolation at parameter $t \in [0, 1]$ is:

$$\text{slerp}(\mathbf{A}, \mathbf{B}, t) = \frac{\sin((1-t)\theta)}{\sin\theta} \mathbf{A} + \frac{\sin(t\theta)}{\sin\theta} \mathbf{B}, \tag{4}$$

where $\theta = \arccos(\mathbf{A} \cdot \mathbf{B})$. Every point on this arc is a unit octonion with $\mathcal{H} = 0$. Linear interpolation (LERP) passes through the interior of S^7, violating the unit-norm constraint and destroying algebraic coherence.

2.3 Model Merging Prior Work

Model Soups [Wortsman et al., 2022a]: arithmetic averaging of models fine-tuned from the same checkpoint with different hyperparameters. **Task Arithmetic** [Ilharco et al., 2023]: task vectors defined as weight differences, combined by addition in float space. **TIES-Merging** [Yadav et al., 2023]: resolves sign conflicts between task vectors before merging. **DARE** [Yu et al., 2023]: random weight dropping before merging. **WiSE-FT** [Wortsman et al., 2022b]: linear interpolation between fine-tuned and pretrained weights.

All prior work operates in float weight space with arithmetic operations. RF is the first method using S^7 as the native merging manifold.

3 Method

3.1 The Native 8-Channel Property

A convolutional layer with weight tensor $\mathbf{W} \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k_H \times k_W}$ where $C_{\text{in}} \equiv 0 \pmod 8$ can be reshaped into $N = C_{\text{out}} \cdot k_H \cdot k_W \cdot (C_{\text{in}}/8)$ groups of 8 consecutive input-channel weights:

$$\mathbf{K} = \text{reshape}(\mathbf{W}) \in \mathbb{R}^{N \times 8}. \tag{5}$$

Each row of $\mathbf{K}$ is a natural 8D vector. The RF fold normalizes each group onto S^7:

$$\mathbf{O} = \frac{\mathbf{K}}{\|\mathbf{K}\|}, \quad \boldsymbol{\nu} = \|\mathbf{K}\| \in \mathbb{R}^{N \times 1}. \tag{6}$$

Reconstruction is exact by construction:

$$\hat{\mathbf{W}} = \text{reshape}^{-1}(\mathbf{O} \odot \boldsymbol{\nu}) = \mathbf{W}, \quad \cos(\mathbf{W}, \hat{\mathbf{W}}) = 1.0000. \tag{7}$$

No encoder, no decoder, no learning. The fold is a deterministic projection; the unfold is its exact inverse.

3.2 OctConvNet Architecture

We introduce **OctConvNet**, a convolutional architecture family where every conv layer satisfies $C_{\text{in}} \equiv 0 \pmod 8$.

Oct-aware initialization. For aligned layers, each 8-tuple of input-channel weights is initialized as a near-unit octonion scaled by the fan-in factor:

$$\mathbf{g} = \frac{\mathbf{r}}{\|\mathbf{r}\|} \cdot \sqrt{\frac{2}{C_{\text{in}} \cdot k_H \cdot k_W}}, \quad \mathbf{r} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_8). \tag{8}$$

This places weights near S^7 at initialization.

Table 1: OctConvNet architecture family. All intermediate conv layers satisfy $C \equiv 0 \pmod 8$. The input projection (RGB $\to C_0$) uses standard Kaiming initialization.

Size	Channels	Parameters	Oct kernel groups
S	$8 \to 16 \to 32 \to 64$	$\sim$300K	$\sim$9K
M	$8 \to 32 \to 64 \to 128$	423K	36K
L	$8 \to 64 \to 128 \to 256$	1.68M	143K

Architecture family.

Compatibility with standard architectures. Standard ResNet-18 uses channel widths $[64, 128, 256, 512]$—all multiples of 8. This means *no channel modifications are required* to apply RF to ResNet-18. We introduce **OctResNet-18**, which differs from standard ResNet-18 only in: (1) a 3×3 stem for CIFAR inputs (standard practice), and (2) oct-aware weight initialization. All 19 non-stem conv layers are natively oct-foldable. OctResNet-18 achieves 93.31% on CIFAR-10 with lossless RF fold across 1.39M octonion kernel groups—39× the scale of OctConvNet-M.

3.3 Oct-SLERP Merge

Given two trained OctConvNet checkpoints $\mathcal{A}$ and $\mathcal{B}$ and interpolation parameter $t \in [0, 1]$:

Algorithm 1 OCT-SLERP Model Merge

Require: Models $\mathcal{A}$, $\mathcal{B}$; parameter $t \in [0, 1]$
for each oct-aligned conv layer ℓ **do**
  Extract: $(\mathbf{O}_\ell^{\mathcal{A}}, \boldsymbol{\nu}_\ell^{\mathcal{A}}) \leftarrow \text{fold}(W_\ell^{\mathcal{A}})$
  Extract: $(\mathbf{O}_\ell^{\mathcal{B}}, \boldsymbol{\nu}_\ell^{\mathcal{B}}) \leftarrow \text{fold}(W_\ell^{\mathcal{B}})$
  SLERP: $\mathbf{O}_\ell^M \leftarrow \text{slerp}(\mathbf{O}_\ell^{\mathcal{A}}, \mathbf{O}_\ell^{\mathcal{B}}, t)$ ▷ Eq. (4)
  Interpolate norms: $\boldsymbol{\nu}_\ell^M \leftarrow (1-t)\,\boldsymbol{\nu}_\ell^{\mathcal{A}} + t\,\boldsymbol{\nu}_\ell^{\mathcal{B}}$
  Reconstruct: $W_\ell^M \leftarrow \text{unfold}(\mathbf{O}_\ell^M, \boldsymbol{\nu}_\ell^M)$ ▷ Eq. (7)
end for
for all other layers (BN, head, proj) **do**
  Float interpolate: $W^M \leftarrow (1-t)\,W^{\mathcal{A}} + t\,W^{\mathcal{B}}$
end for
return merged model $\mathcal{M}$; $\mathcal{H}(\mathcal{M}) = 0.000000$

The merged model's oct layers satisfy $\mathcal{H} = 0$ exactly, because SLERP on S^7 never leaves S^7.

3.4 Task Vectors in Oct-Space

The logarithmic map at base model $\mathcal{A}$ toward fine-tuned model $\mathcal{B}$ defines a tangent-space task vector:

$$\mathbf{v} = \frac{\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{A}}{\|\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{A}\|} \cdot \arccos(\mathbf{A} \cdot \mathbf{B}). \tag{9}$$

Applying at scale s via the exponential map:

$$\text{apply}(\mathbf{A}, \mathbf{v}, s) = \cos(s\theta)\,\mathbf{A} + \sin(s\theta)\,\frac{\mathbf{v}}{\|\mathbf{v}\|}. \tag{10}$$

At $s{=}1$: recovers $\mathcal{B}$ exactly. At $s{=}{-1}$: anti-task (unlearning direction). This is the oct-space analogue of task arithmetic [Ilharco et al., 2023].

4 Experiments

4.1 Setup

Hardware. Single NVIDIA GTX 1660 Ti (6 GB VRAM), CUDA 12.1, PyTorch 2.5.1.

Training. AdamW, OneCycleLR (max_lr$= 3 \times 10^{-3}$), weight decay 10^{-4}, label smoothing 0.1, batch size 128. Augmentation: random crop (padding $= 4$), horizontal flip, color jitter.

Datasets. CIFAR-10 and CIFAR-100 [Krizhevsky, 2009] ($32{\times}32$ images, 50K/10K train/test splits).

Architecture. OctConvNet-M unless stated. Float32 baseline uses identical architecture with standard Kaiming initialization.

4.2 RF Fold Losslessness

We train OctConvNet-M and OctConvNet-L on CIFAR-10 and CIFAR-100, then apply the RF fold to all conv layers simultaneously.

The lossless property holds after any training run and is independent of dataset difficulty, model scale, and architecture family. Notably, OctResNet-18—a standard ResNet-18 requiring no channel modifications—achieves lossless fold across all 19 eligible layers and $1{,}394{,}688$ octonion kernel groups (93.31% CIFAR-10 accuracy), $39\times$ the scale of OctConvNet-M.

4.3 OctConvNet vs. Float32 Baseline

4.4 Oct-SLERP Merging: Constructive Divergence

We train a base OctConvNet-M (88.45%) then produce two fine-tuned variants using complementary augmentation strategies: **Fine-tune A** (texture): high contrast and sharpening emphasis, specializing in high-frequency texture cues.

Table 2: RF fold is lossless across all models, datasets, and scales. Holo values at $\sim 9\times 10^{-16}$ are at float32 machine epsilon.

Model	Dataset	Layers	Oct groups	$\cos(\hat{\mathbf{W}}, \mathbf{W})$	$\mathcal{H}$
OctConvNet-M	CIFAR-10	6	36 000	1.000000	0.000000
OctConvNet-M	CIFAR-100	6	36 000	1.000000	9.26×10^{-16}
OctConvNet-L	CIFAR-10	6	143 424	1.000000	0.000000
OctResNet-18	CIFAR-10	19	1 394 688	1.000000	9.40×10^{-16}

Table 3: OctConvNet-M matches or exceeds float32 accuracy on CIFAR-10 at identical parameter count (423 026 parameters, 30 epochs).

Model	Accuracy	Δ
Float32 ConvNet-M (Kaiming init)	88.15%	—
OctConvNet-M (oct-init)	88.34%	+0.19%

Fine-tune B (shape): blur simulation, grayscale conversion, and large crops, specializing in low-frequency global shape cues.

Table 4: Oct-SLERP outperforms both float averaging and the best individual fine-tune across all runs and datasets. $\mathcal{H} = 0.000000$ at every merge point in all experiments. Best t found by validation sweep. †: original run.

Dataset	Seed	Best indiv.	Float avg	Best OCT-SLERP	t^*	Δbest	Δfloat
CIFAR-10†	42	88.69%	88.45%	**88.88%**	0.200	**+0.19%**	+0.43%
CIFAR-10	123	88.29%	89.27%	**89.35%**	0.025	**+0.06%**	+0.08%
CIFAR-100	42	62.02%	62.08%	**62.42%**	0.225	**+0.40%**	+0.34%
CIFAR-10 mean ± std (seeds 42, 123)						+0.23% ± 0.24%	
OctResNet-18‡	CIFAR-10	93.32%	93.96%	**93.67%**	0.325	**+0.35%**	−0.29%

‡ ResNet-18 architecture (11.2M params). Float avg benefits here (same-task fine-tunes); SLERP still outperforms both individual models (+0.35%). See Section 5.

The optimal t varies by run (0.025–0.225) and reflects the relative magnitude of directional displacement in weight space by each specialist. Across all three conditions, the geodesic arc on S^7 accesses weight-space regions unavailable to either individual training trajectory: SLERP beats the best individual model by +0.19% (CIFAR-10, seed 42), +0.06% (CIFAR-10, seed 123), and +0.40% (CIFAR-100, seed 42). The CIFAR-100 advantage is the strongest—with 100 classes, the texture and shape specialists diverge more meaningfully. On CIFAR-100, SLERP outperforms the best individual model across 21 consecutive sweep points ($t = 0.025$ through $t = 0.500$).

4.5 Oct-SLERP Merging: Complementary Splits

We split the training set in half (25K/25K) and train two OctConvNet-M models independently, one on each split.

The characteristic W-shaped sweep curve (high near endpoints, collapsed near t=0.5) is consistent across both datasets. Float averaging hits a geometric cancellation zone; OCT-SLERP avoids it.

Note. The best OCT-SLERP result here is t=0 or t=1 (one of the original models), because neither model benefits from the other's held-out data. This protocol demonstrates *robustness*, not superiority over individual models. Section 4.4 is the correct primary claim.

4.6 Scale Invariance

4.7 Holographic Coherence Across All Experiments

Holographic coherence was measured at every OCT-SLERP interpolation point across all experiments. Total measurements: 123.

Table 5: Float averaging collapses to random chance on both datasets. OCT-SLERP degrades gracefully and recovers symmetrically, demonstrating S^7 robustness under extreme complementarity.

Method	CIFAR-10	CIFAR-100
Model A (split 1)	85.39%	52.00%
Model B (split 2)	85.27%	52.33%
Float average	10.00%	0.99%
OCT-SLERP t=0.10	83.75%	46.38%
OCT-SLERP t=0.25	50.46%	9.56%
OCT-SLERP t=0.90	82.09%	48.24%
OCT-SLERP best	85.39%	52.33%
SLERP vs. float avg	+75.39%	+51.34%

Table 6: The RF fold and OCT-SLERP advantage are scale-invariant. $4\times$ model scale produces ± 0.004 change in SLERP advantage.

Metric	OctConvNet-M	OctConvNet-L	Δ
Parameters	423K	1.68M	$4.0\times$
Oct kernel groups	36 000	143 424	$4.0\times$
RF fold $\cos$	1.000000	1.000000	0.000000
RF fold $\mathcal{H}$	0.00	0.00	0.00
Best SLERP Δ vs. float avg	+0.4198	+0.4160	−0.0038

Result: $\mathcal{H} = 0.000000$ **at all 123 measurement points.** The algebraic structure of S^7 is maintained exactly throughout every geodesic arc between any two OctConvNet checkpoints.

5 Analysis

Why SLERP outperforms float averaging. Float averaging computes $(\mathbf{A} + \mathbf{B})/2$, the arithmetic midpoint, which lies *inside* S^7. For weight vectors far apart in angle, this midpoint is near the origin—producing near-zero weights that represent neither training distribution. OCT-SLERP computes the midpoint *on* the sphere surface, preserving magnitude and computing a geometrically correct directional average. For constructive divergence, the optimal $t<0.5$ accesses a weight-space region that neither individual model explores.

The optimal t and task asymmetry. The optimal t=0.20 means the texture specialist contributes 80% of the geodesic distance and the shape specialist 20%. This asymmetry reflects that texture specialization under our augmentation strategy produced a larger and more useful directional displacement in weight space. The optimal t is a continuous hyperparameter that can be found by sweep on a validation set using the provided `rf-merge -sweep` CLI.

ResNet-18 results and the role of task divergence. On OctResNet-18 with same-task fine-tunes (both specialists trained on CIFAR-10), float averaging scores 93.96%—higher than either individual model (93.32%). This is the Model Soups effect: when fine-tunes are near-identical in task, arithmetic averaging benefits from implicit ensembling. OCT-SLERP at t=0.325 scores 93.67%, outperforming both individual models (+0.35%) but not float averaging (−0.29%).

This result is not a failure—it reveals the operating regimes of each method. Float averaging performs well when models are similar; OCT-SLERP outperforms individual models in all tested conditions and outperforms float averaging when tasks are divergent (constructive divergence: +0.43%; complementary split: +75.4%). The choice of merge strategy should match the degree of task divergence between models being merged.

Comparison with float-space methods. Float averaging, Task Arithmetic [Ilharco et al., 2023], TIES-Merging [Yadav et al., 2023], and DARE [Yu et al., 2023] all operate in float weight space with no manifold constraint. RF provides a geometric constraint—all merged weights are guaranteed to remain on S^7 with $\mathcal{H} = 0$ exactly—that float-space methods lack.

Limitations. **Architecture constraint.** RF requires networks with 8-aligned channel counts. Adapting standard architectures requires re-training from scratch. **LLM weights.** We tested RF on DistilGPT-2 (82M parameters) and found catastrophic degradation (KL $= 7.1$, PPL $+28{,}375\%$). LLM attention matrices have global coupling that defeats 8D encoding. RF is not a general-purpose compression scheme. **Margin.** The improvement over best individual is $+0.19\%$ (CIFAR-10) and $+0.40\%$ (CIFAR-100). We expect larger advantages at greater task divergence.

6 Related Work

Hypercomplex neural networks. Deep Octonion Networks [Wu et al., 2023] use octonion-valued activations during forward passes. Quaternion NNs [Parcollet et al., 2019] similarly use 4D algebraic structure as a computational primitive. The key distinction: these methods use hypercomplex numbers as the *computational* algebra during forward passes; RF uses S^7 specifically as a *weight representation* for merging—the OctConvNet forward pass is standard float32 convolution.

Model merging. Wortsman et al. [2022a], Ilharco et al. [2023], Yadav et al. [2023], Yu et al. [2023] provide the primary baselines, all operating in float weight space. WiSE-FT [Wortsman et al., 2022b] applies linear interpolation between fine-tuned and pretrained weights—the closest conceptual precursor to OCT-SLERP, but using LERP rather than geodesic interpolation.

Riemannian model averaging. Riemannian mean computation on manifolds of weight matrices [Bhatia, 2007] provides a theoretical framework for manifold-aware averaging but has not been applied to neural network merging at scale. RF provides a practical instantiation via S^7.

7 Continual Learning via Sequential SLERP

A natural extension of OCT-SLERP merging is continual learning: rather than merging two independently trained models, we sequentially merge a model's new task specialization back toward the previous checkpoint after each task. We call this **SLERP-CL**.

Protocol. We train a base OctResNet-18 on all 10 CIFAR-10 classes, then evaluate three continual learning methods on a two-task split: Task A (classes 0–4: airplane, automobile, bird, cat, deer) and Task B (classes 5–9: dog, frog, horse, ship, truck).

Methods compared. **Naive** sequential fine-tuning serves as the forgetting baseline. **EWC** [Kirkpatrick et al., 2017] adds a Fisher-information penalty to slow weight changes important for Task A during Task B training. **SLERP-CL** (ours): after fine-tuning on each task, merge the new specialist back toward the previous checkpoint via OCT-SLERP at t=0.3, then use the merged checkpoint as the starting point for the next task.

Table 7: Continual learning results on CIFAR-10 two-task split. Forgetting = Task A accuracy drop after Task B training. SLERP-CL reduces forgetting by 92% vs. naive and outperforms EWC without any Fisher matrix computation.

Method	Task A	Task B	All	Forgetting
Base model	91.04%	93.94%	92.49%	—
Naive fine-tuning	0.00%	97.92%	48.96%	−95.56%
EWC [Kirkpatrick et al., 2017]	0.00%	98.16%	49.08%	−95.86%
SLERP-CL (ours, t=0.3)	**87.04%**	**95.92%**	**91.48%**	**−8.04%**

Results. Naive fine-tuning and EWC both collapse Task A to 0% accuracy — the model's output layer becomes entirely Task-B-specific and the backbone weights shift to accommodate the new class set. SLERP-CL retains 87.04% on Task A while achieving 95.92% on Task B, reducing forgetting from 95.56% to 8.04% ($\times 11.9$ improvement). Overall accuracy on all 10 classes (91.48%) nearly matches the base model (92.49%), compared to $\sim 49\%$ for the catastrophic forgetting baselines.

Why EWC fails here. EWC's Fisher penalty is computed on Task A samples and penalises changes to weights with high gradient variance on Task A. However, when Task A and Task B require entirely different output logits (the final linear layer maps 5 classes vs. 5 different classes), the penalty cannot prevent the output layer from overwriting Task A representations. SLERP-CL bypasses this limitation by operating geometrically: after each task, it pulls the new specialist back toward the previous checkpoint on S^7, preserving prior directional structure in weight space regardless of which layers changed.

Computational cost. EWC requires one full forward+backward pass over Task A data to compute the Fisher diagonal ($\sim N$ gradient evaluations). SLERP-CL requires a single fold+merge operation: $O(G)$ where G is the number of octonion kernel groups (1.39M for OctResNet-18). No gradient computation is needed. At inference time both methods are identical.

7.1 Partial-Network LLM Merging

A natural question is whether Resonance Folding extends to transformer-based language models. Global attention matrices resist local 8D folding due to cross-position coupling, but feed-forward MLP blocks are structurally analogous to position-wise 1×1 convolutions: each token is processed independently, making the 8D local grouping geometrically meaningful.

Fold losslessness on SmolLM2-135M. We apply RF fold to all 90 MLP weight layers of SmolLM2-135M (LLaMA-style: `gate_proj`, `up_proj`, `down_proj`). SmolLM2-135M contains 134.5M parameters, of which 79.6M (59%) reside in MLP blocks. All 90 layers fold losslessly: $\cos = 1.000000$, $\mathcal{H} = 7.96 \times 10^{-16}$, across 9,953,280 octonion kernel groups.

Partial-network SLERP merge. We merge SmolLM2-135M (Base) and SmolLM2-135M-Instruct: OCT-SLERP on the 90 MLP layers, float interpolation on the 183 remaining layers.

Table 8: Partial-network merge perplexity on WikiText-2 test set. OCT-SLERP at t=0.3 outperforms float averaging by 0.40 PPL. $\mathcal{H} = 1.14 \times 10^{-16}$ at all merge points.

Method	PPL (↓)	Δ vs float avg	$\mathcal{H}$
Base model (SmolLM2-135M)	19.95	—	—
Instruct model	22.91	—	—
Float average (t=0.5)	20.49	baseline	—
OCT-SLERP t=0.3	**20.09**	$\mathbf{-0.40}$	1.14×10^{-16}
OCT-SLERP t=0.5	20.49	+0.00	1.14×10^{-16}
OCT-SLERP t=0.7	21.19	+0.70	1.14×10^{-16}

OCT-SLERP at t=0.3 achieves lower perplexity than float averaging while preserving $\mathcal{H} = 0$ exactly. The instruct model's higher raw perplexity is expected — instruction tuning trades token-level prediction for conversational alignment. MLP-only folding applies to any transformer architecture where MLP projection dimensions are multiples of 8; attention projections are float-averaged.

8 Conclusion

We have introduced Resonance Folding, a weight representation technique that projects native 8-channel convolutional filter weights onto S^7 using the Fano-plane octonion algebra. The key properties—losslessness, holographic coherence

preservation, and OCT-SLERP merging advantage—are all empirically verified across multiple scales, datasets, and experimental conditions.

The central result: when two OctConvNet models are fine-tuned with complementary specializations, geodesic OCT-SLERP interpolation at $t{=}0.20$ produces a merged model that outperforms both individual fine-tunes ($+0.19\%$) and their arithmetic average ($+0.43\%$). The algebraic structure of S^7 is doing real work—it is not a re-parameterization but a constraint that guides interpolation toward a geometrically coherent combined representation unavailable to either individual training trajectory.

Future work includes extending SLERP-CL to longer task sequences (beyond two tasks), applying RF to transformer architectures with 8-aligned attention heads, and systematic comparison against TIES-Merging and DARE on standard NLP benchmarks.

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